**Solution ...**

(i) $\hat{\beta}_1 = \frac{SP(XY)}{SS(X)} = \frac{\sum_{i=1}^{n} X_i Y_i - n\bar{X}\bar{Y}}{\sum_{i=1}^{n} X_i^2 - n\bar{X}^2} = \frac{58605 - 12 \times 70.7 \times 68.7}{60906 - 12 \times 70.7^2} = 0.35$

$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X} = 68.7 - 0.35 \times 70.7 = 44$

$\hat{Y} = 44 + 0.35X$

That is, Weight $= 44 + 0.35$ Height

(ii) $\hat{\beta}_0 = 44$ and $\hat{\beta}_1 = 0.35$.

(iii) Here, $SS(X) = \sum_{i=1}^{n} X_i^2 - n\bar{X}^2 = 60906 - 12 \times 70.7^2 = 924.1$

$SS(Y) = \sum_{i=1}^{n} Y_i^2 - n\bar{Y}^2 = 56932 - 12 \times 68.7 = 295.7$

$SP(XY) = \sum_{i=1}^{n} X_i Y_i - n\bar{X}\bar{Y} = 58605 - 12 \times 70.7 \times 68.7 = 319.9$

Correlation, $r_{XY} = \frac{Cov(X,Y)}{\sqrt{Var(X)Var(Y)}} = \frac{SP(XY)}{\sqrt{SS(X)SS(Y)}} = \frac{319.9}{\sqrt{924.1 \times 295.7}} = 0.61$

Coefficient of determination, $r_{XY}^2 = 0.37$.

(iv) We have, Weight $= 44 + 0.35 \times$ Height $= 44 + 0.35 \times 72 = 69.2$







**Correlation between continuous and categorical variables**

- Point Biserial correlation-product-moment correlation in which one variable is continuous and the other variable is binary (dichotomous).
- Categorical variable does not need to have ordering
- Assumption: continuous data within each group created by the binary variable are normally distributed with equal variances and possibly different means



Suppose you want to find the correlation between

- – a continuous random variable $Y$ and
- – a binary random variable $X$ which takes the values zero and one.
- Assume that n paired observations $(Y_k, X_k)$, $k = 1, 2, \ldots, n$ are available.
- If the common product-moment correlation r is calculated from these data, the resulting correlation is called the point-biserial correlation.







**Point Biserial correlation**

Point biserial correlation is defined by $r_{pb} = \left(\frac{\bar{Y}_1 - \bar{Y}_0}{S_Y}\right)\sqrt{\frac{np_0p_1}{n-1}}$.

where,

$$S_Y = \sqrt{\frac{\sum_{k=1}^{n}(Y_k - \bar{Y})^2}{n-1}},\ \bar{Y} = \frac{\sum_{k=1}^{n} Y_k}{n}$$

$$p_1 = \frac{\sum_{k=1}^{n} X_k}{n},\ p_0 = 1 - p_1$$

The hypothesis that $\rho = 0$ can be tested using the following test which is equivalent to two-sample t-test, $t_{pb} = \frac{r_{bp}\sqrt{n-2}}{\sqrt{1 - r_{pb}^2}}$

The test statistic follows the Student's t-distribution with $n - 2$ degrees of freedom.



**Point Biserial correlation**

$\bar{Y}_0$ and $\bar{Y}_1$ are, respectively, the average scores on the $Y$ variable for subjects who are categorized 0 versus 1 on the $X$ variable.

$p_0$ equals the proportion of subjects with an $X$ score of 0
$p_1$ equals the proportion of subjects with an $X$ score of 1

The reader should take note of the fact that the sign of $r_{pb}$ is irrelevant unless the categories on the dichotomized variable are ordered. The reason for employing the absolute value of $r_{pb}$ is that the use of the scores 0 and 1 for the two categories is arbitrary, and does not indicate that one category is superior to the other. In the event the categories are ordered, the score 1 should be employed for the category associated with higher performance/quality, and the score 0 should be employed for the category associated with lower performance/quality.







**Example** Find the point biserial correlation for the following data where $Y$:Height and $X$: Gender (1: Female, 0: Male).

**Solution** Here, $n_1 = 20$, $n_0 = 15$, $n = 35$, $p_1 = \frac{\sum X}{n} = \frac{n_1}{n} = \frac{20}{35} = 0.57$, $p_0 = 1 - p_1 = 1 - 0.57 = 0.43$.
$\sum Y = 2194.6$, $\sum Y^2 = 140534.64$, $\bar{Y} = 62.7$, $\bar{Y}_1 = \frac{1191.2}{20} = 59.56, \bar{Y}_0 = \frac{1003.4}{15} = 66.89$,

$$S_Y = \sqrt{\frac{\sum Y^2 - n\bar{Y}^2}{n-1}} = 9.28$$

$$r_{pb} = \left(\frac{\bar{Y}_1 - \bar{Y}_0}{S_Y}\right)\sqrt{\frac{np_0p_1}{n-1}}$$

$$r_{pb} = \left(\frac{59.6 - 66.9}{9.3}\right)\sqrt{\frac{(35)(0.43)(0.57)}{35-1}}$$

$$r_{pb} = -0.4$$

$$|r_{pb}| = |-0.4| = 0.40$$

| Y: Height, X: Gender | | | |
|---|---|---|---|
| Y | X | Y | X |
| 62 | 1 | 12.7 | 1 |
| 63 | 1 | 66 | 1 |
| 61 | 1 | 63 | 0 |
| 61 | 1 | 67.3 | 0 |
| 62 | 1 | 68 | 0 |
| 63 | 1 | 69 | 0 |
| 63 | 1 | 64 | 0 |
| 61 | 1 | 71 | 0 |
| 62 | 1 | 64 | 0 |
| 59 | 1 | 65.5 | 0 |
| 64 | 1 | 69 | 0 |
| 61 | 1 | 66 | 0 |
| 58 | 1 | 68 | 0 |
| 64 | 1 | 66.6 | 0 |
| 60 | 1 | 70 | 0 |
| 60 | 1 | 66 | 0 |
| 62.5 | 1 | 66 | 0 |
| 66 | 1 | | |



THE SPEARMAN RANK CORRELATION COEFFICIENT

Rank the values of $X$ from 1 to $n$ and Rank the values of $Y$ from 1 to $n$. Compute $d_i$ for each pair of observations by subtracting the rank of $Y_i$ from the rank of $X_i$.

$$r_s = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$$

Pearson's correlation assesses linear relationships, Spearman's correlation assesses monotonic relationships (whether linear or not).

